

ARTIFICIAL INTELLIGENCE IN THE FINANCE DOMAIN: A STUDY OF AI-DRIVEN FRAUD DETECTION AT MASTERCARD (DECISION INTELLIGENCE)

Domain: Finance

AI Application Selected: Mastercard Decision Intelligence & Decision Intelligence Pro

Name: Yash Sharma
Reg no: 192525150
Course code: MLA0104
Course name: AIES

INTRODUCTION AND BACKGROUND

Every time a card is swiped, tapped or entered online, a payment network has a few hundred milliseconds to decide whether the transaction should go through. Mastercard processes this decision for a very large share of the world's card transactions, and for years it has relied on a system called Decision Intelligence to do this. DI does not simply check a transaction against a blacklist; it generates a risk score between 0 and 999 for practically every transaction that passes through the network, using patterns learned from historical transaction data, network-wide fraud signals and behavioural indicators.

The system was originally introduced to move card networks away from static, rule-based fraud checks (for example, “decline if amount > \$2000 in a new country”) toward adaptive, learning-based models that update as fraud patterns shift. In 2019 Mastercard significantly expanded DI with deep learning components, and it has continued to add capability since, most notably an in-house recurrent neural network branded Decision Intelligence Pro, which is trained on historical transaction sequences across the network to address weaknesses in earlier-generation fraud models, particularly their tendency to wrongly decline genuine purchases.

This report looks at why such a system is needed, what AI techniques power it, how data moves through it, and where it delivers value as well as where it still runs into difficulty.

PROBLEM IDENTIFICATION

The real-world problem is straightforward to state but hard to solve at scale: card fraud costs the payments industry tens of billions of dollars every year, and traditional rule-based fraud systems are not good enough to stop it without also blocking a large number of legitimate customers. Card fraud losses across the industry are estimated in the tens of billions of dollars annually, and the knock-on cost of false declines (genuine transactions wrongly rejected) is estimated to run into the hundreds of billions of dollars a year in lost merchant revenue, because a meaningful share of shoppers simply abandon a merchant after being wrongly declined once.

Older systems used fixed thresholds and manually written rules. Fraudsters adapted quickly to these fixed rules, and the rules themselves could not keep pace with fast-changing tactics such as synthetic identities, card-skimming variants, and coordinated “fraud rings” that spread small transactions across many merchants and cards to avoid detection. The core challenge, then, is to catch a rapidly-evolving, low-frequency but high-cost pattern (fraud) hidden inside an enormous volume of legitimate activity, in real time, without materially slowing down or annoying genuine customers.

NEED FOR AI SOLUTION

Rule-based systems struggle with this problem for three structural reasons. First, they cannot scale to the number of variables that matter in a modern transaction. Mastercard's own description of DI states that the system evaluates several hundred distinct data points per transaction, ranging from transaction context to device fingerprint to merchant relationship history — far beyond what a human-authored rule set can realistically encode. Second, rules are static, while fraud tactics are not; a rule tuned to catch last year's fraud pattern is often blind to this year's. Third, rule-based decline thresholds are blunt instruments — they cannot weigh dozens of weak, individually inconclusive signals together the way a trained model can.

Machine learning and deep learning address exactly this gap. Instead of hand-written thresholds, models trained on hundreds of millions of past transactions learn the subtle, non-linear combinations of signals that separate fraud from normal behaviour, and they can be retrained as new fraud patterns emerge. This is why AI, rather than traditional software logic, is the appropriate solution: the problem is fundamentally a pattern-recognition and adaptive-learning problem operating on data volumes and speeds that manual rule design cannot match.

AI TECHNOLOGIES USED

Decision Intelligence combines several AI techniques rather than relying on a single model type:

- Deep Neural Networks: Multi-layered networks progressively transform raw transaction fields into higher-level behavioural risk signals, capturing non-linear relationships across hundreds of variables that simple rule logic cannot represent.
- Recurrent Neural Networks (RNN): Decision Intelligence Pro uses an RNN architecture trained on historical sequences of transactions (Mastercard cites training on roughly 125 billion transactions a year across its network), allowing the model to reason about a cardholder's spending pattern over time rather than looking at one transaction in isolation.
- Graph-based relationship analysis: The system looks beyond a single merchant or cardholder to assess relationships across multiple entities — cards, devices, merchants — which helps expose coordinated fraud rings that would look innocuous transaction-by-transaction.
- Behavioural biometrics and device intelligence: Device fingerprints, browser/OS signals and interaction patterns feed into the risk score as an additional, harder-to-spoof layer of evidence.
- Class-imbalance handling (e.g. SMOTE-style techniques): Because genuine transactions vastly outnumber fraudulent ones, specialised sampling and weighting techniques are used so the model does not simply learn to always predict “not fraud.”

- Explainable AI (XAI) methods such as SHAP/LIME-style attribution: Increasingly required so that a flagged transaction can be traced back to the specific factors that drove the score, both for analyst review and for regulatory justification.

WORKING MECHANISM

At a high level, Decision Intelligence works in five stages, shown in the architecture diagram below. Real-time transaction, device and network data is ingested and immediately pre-processed — cleaned, normalised, and enriched with computed features such as velocity counts (how often this card/device has transacted recently). This engineered data is then passed into the model core, where a recurrent neural network evaluates the sequential spending pattern of the cardholder, a graph-based component checks relationships between the card, merchant and device against known fraud clusters, and an ensemble scoring layer merges these signals into a single risk score from 0 to 999.

Before that score reaches a decision point, an explainability layer attaches a human-readable reason (for example, “unusual device + high velocity + new merchant category”) to any transaction above a risk threshold, which supports both fraud-analyst review and regulatory audit trails. Finally, the score and its explanation feed a decisioning step: very low-risk transactions are auto-approved, very high-risk ones are auto-declined, and borderline cases are routed to a human fraud analyst, with the final decision returned to the issuer's authorization system — all of this typically completing within a few hundred milliseconds so it does not add noticeable delay at checkout.

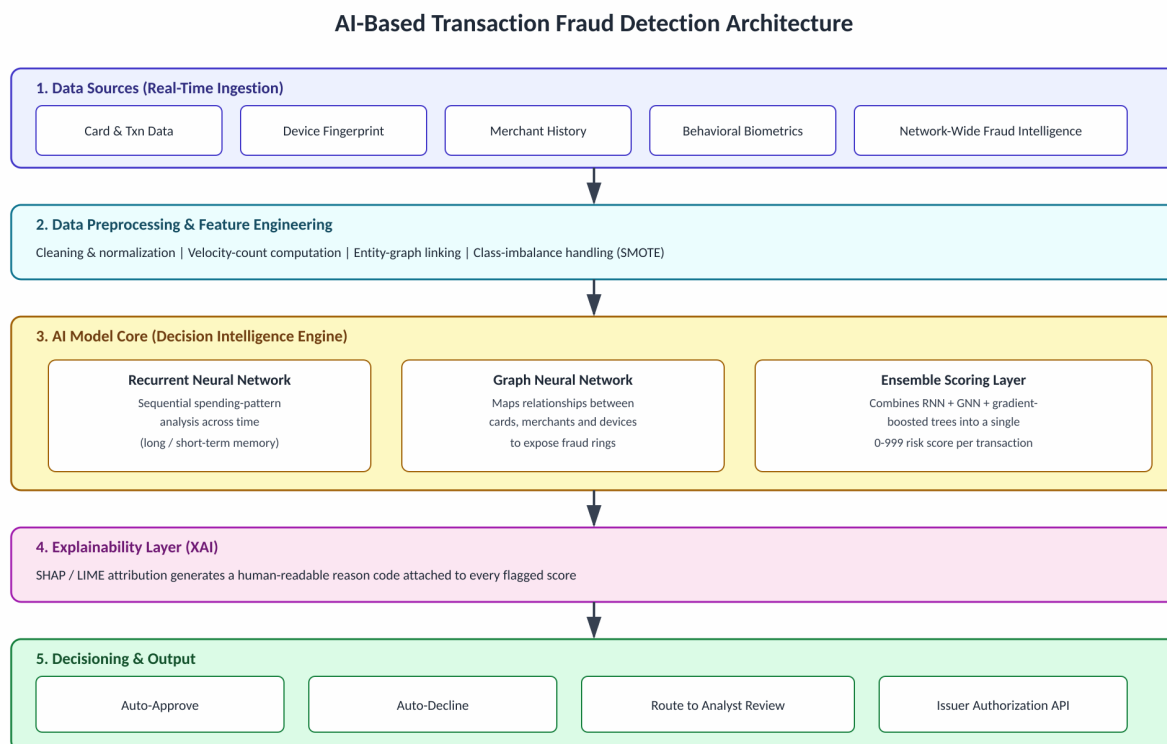


Figure 1: Layered architecture of the AI-based fraud detection pipeline

DATA SOURCES AND PROCESSING

The system draws on several categories of data. Transaction-level data covers amount, merchant category code, geolocation, time of day and payment channel. Device and behavioural data covers device fingerprints, browser/OS characteristics and interaction patterns (behavioural biometrics). Historical and network data includes each cardholder's own transaction history and, importantly, Mastercard's proprietary network-wide fraud intelligence — patterns observed across its entire card network rather than just one issuer's portfolio. Merchant data covers a merchant's own historical fraud/chargeback rates.

This data is processed in real time: it is cleaned and normalised, engineered into features such as velocity counts and entity-relationship graphs, and rebalanced to correct for the fact that fraudulent transactions are a tiny minority of all transactions. The processed features are what the neural network and ensemble layers actually consume when producing a score for intelligent, near-instant decision-making.

REAL-WORLD IMPLEMENTATION

Decision Intelligence is embedded directly in the authorization path of Mastercard's payment network, so it is used by banks and card issuers, payment gateways, and merchants (particularly those in high-risk verticals) around the world, essentially anywhere a Mastercard-branded transaction is processed. Issuers use the risk score and reason codes to set their own approve/decline rules and thresholds, while fraud analysts use the explainability output to investigate borderline cases flagged by the model rather than reviewing transactions blind.

Because DI operates at the network level rather than being installed by one bank alone, it benefits every participating issuer from patterns learned across the whole network — a fraud pattern first seen at one institution can help protect a completely different institution within the same scoring update cycle.

IMPACT AND BENEFITS

- **Reduced false declines:** Banks that implement AI fraud detection with explainability built in, backed by proper validation, report 40–60% reductions in false positives compared with legacy rule-based systems, directly protecting merchant revenue and customer trust.
- **Faster, near-instant decisions:** Risk scoring completes within the authorization window (a few hundred milliseconds), so fraud prevention does not add friction at checkout, and a first production deployment for alert triage can typically go live within 90–120 days.
- **Better detection of coordinated fraud:** Graph-based relationship analysis catches fraud rings and synthetic-identity clusters that transaction-by-transaction rule checks would miss.
- **Cost savings for issuers:** Institutions moving from rule-based tools to AI models report over 50% reductions in manual review costs alongside meaningfully better suspicious-activity detection.
- **Improved customer experience:** 83% of surveyed institutions report that AI has significantly reduced false positives and customer churn, since fewer genuine purchases are wrongly declined.

CHALLENGES AND LIMITATIONS

- Explainability / the “black-box” problem: Deep learning models are inherently harder to interpret than rule-based systems, yet regulators require institutions to justify why a specific transaction was flagged, particularly for Suspicious Activity Reports under Bank Secrecy Act-style requirements.
- Regulatory and fairness scrutiny: Financial regulators have flagged explainability/transparency and bias/discrimination as the most pressing AI concerns in banking, since a model that disproportionately flags certain customer segments creates legal exposure even without explicit demographic inputs.
- Data quality and class imbalance: Fraudulent transactions are a very small fraction of total volume, which biases naive models toward always predicting “legitimate” unless carefully corrected.
- Privacy versus effectiveness trade-off: Better fraud detection requires access to more customer and behavioural data, which creates tension with data-privacy expectations and regulation.
- Adversarial adaptation: Fraudsters actively study and probe detection systems, meaning models require continuous retraining rather than one-time deployment.
- Human-AI collaboration overhead: Because full automation of high-risk decisions is not yet trusted for borderline cases, systems must route a share of transactions to human analysts, which still requires investment in analyst tooling and training.

FUTURE SCOPE AND EMERGING TRENDS

- Wider use of explainable AI (XAI) techniques such as SHAP and LIME to turn black-box scores into regulator- and analyst-readable justifications by default, rather than as an add-on.
- Greater use of generative AI to help fraud analysts query large transaction datasets and policy documents in natural language, speeding up investigation and analyst training.
- Deeper graph-based and network-level modelling to catch increasingly coordinated, multi-account fraud rings and synthetic-identity fraud.
- Tighter integration of AI fraud detection with real-time regulatory compliance reporting (for example, Basel III-style capital and liquidity visibility), reducing reporting lag from days to near real time.

- Stronger AI governance frameworks — board-level accountability, bias testing protocols and model risk management extended explicitly to AI — as institutions move from pilot projects to full production deployment.

CONCLUSION

Mastercard's Decision Intelligence illustrates why AI has become central to modern financial security. By replacing static, rule-based fraud checks with adaptive deep learning and graph-based models trained on network-wide data, it addresses a problem that pure rule engines never could: separating a rare but costly fraud signal from an enormous volume of legitimate activity, in real time, without materially inconveniencing genuine customers. The system has delivered measurable benefits in reduced false declines and faster, more accurate decisions.

At the same time, its limitations — particularly around explainability, fairness and the constant need to retrain against adversarial adaptation — show that AI in finance is not a “set and forget” solution. The direction the industry is now moving in, toward explainable AI, stronger governance and closer human-AI collaboration, suggests that the next phase of AI-driven fraud detection will be judged as much on transparency and accountability as on raw detection accuracy.

REFERENCES

1. Mastercard. “Advanced Fraud Detection and Decisioning Tools.” Mastercard Global. mastercard.com/global/en/business/cybersecurity-fraud-prevention/risk-decisioning/fraud-and-decisioning.html
2. Mastercard. “AI is helping banks save millions by transforming payment fraud prevention.” Mastercard Insights, 2026. mastercard.com/us/en/news-and-trends/Insights/2026/ai-is-helping-banks-save-millions-by-transforming-payment-fraud-prevention.html
3. “How Mastercard Uses AI to Stop Payment Fraud in Milliseconds.” CodeSmiths, 2026. codesmiths.in/ai-powered-fraud-prevention-digital-payments/
4. “Artificial Intelligence: Mastercard’s new Neural Network for enhanced fraud detection.” Substack, 2026. lex.substack.com/p/artificial-intelligence-mastercards
5. “AI Fraud Detection in Banking 2026 Guide.” Emburse, 2026. emburse.com/resources/ai-fraud-detection-in-banking

6. "Financial Fraud Detection Using Explainable AI and Stacking Ensemble Methods."
arXiv preprint, 2025. arxiv.org/pdf/2505.10050